DaemonGuard

A GreenOps-Based Intelligent Resource Optimization System for Windows



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Intelligent resource optimisation with safety built in.

- GreenOps

Resource optimisation with energy and carbon awareness.

- Explainable

Recommendations grounded in visible signals and scores.

- Safe

Windows processes remain under user or administrator control.

THE PROBLEM

Windows resource waste is hard to see safely

- Applications, background processes, and services consume CPU, RAM, disk, network, and GPU resources.
- Task Manager exposes usage, but users must decide manually whether a process is necessary or wasting resources.
- Unnecessary resource use can reduce performance and increase energy consumption.

Main question: Can potentially wasteful processes be identified safely while supporting GreenOps-based optimization?

VISIBLE RESOURCE SURFACES

What the system must make legible

CPU

Performance pressure

RAM

Inactive memory

Disk

Background activity

Network

Connection activity

GPU

Application demand

Idle time

Potential waste signal

RESEARCH GAP

Existing tools monitor well, but the research gap remains

EXISTING LANDSCAPE

Windows Task Manager, Microsoft Intune Endpoint Analytics, ControlUp, Nexthink, and Lakeside SysTrack.

Monitoring, analytics, and remediation.

DAEMONGUARD

A research prototype, not a replacement.

Process-level waste analysis and GreenOps-oriented optimisation.

FOCUS

Explainable recommendations and protection of critical Windows processes.

Multi-metric analysis beyond static CPU and RAM thresholds.

Research question: Can multi-metric analysis identify potentially avoidable resource consumption more effectively than static CPU and RAM thresholds?

SYSTEM CONCEPT

DaemonGuard turns raw telemetry into a safe recommendation

01

Windows OS

02

Resource Collector

Reads CPU, RAM, disk, network, GPU, and idle time.

03

Analysis Engine

Calculates a Resource Waste Score and Operational Risk Score.

04

Classification

Healthy, Suspicious, or Potentially Wasteful.

05

Dashboard

Ignore, Optimise, Restart, or Close.

RECOMMENDATION OUTPUT

From telemetry to an informed action

Ignore

Keep a healthy process visible.

Optimize

Highlight potentially wasteful activity.

Restart or Close

Offer a user-controlled next step.

Control stays with the user: the initial version is recommendation-based, not automatic termination.

EXPLAINABLE INTELLIGENCE

A transparent heuristic makes the intelligence explainable

Signal	Score
High memory usage	+3
Long inactivity	+3
Very low CPU usage	+2
No network activity	+1
No disk activity	+1

RESOURCE WASTE SCORE

Classification keeps the recommendation visible and explainable.

0–3	4–6	7–10
Healthy	Suspicious	Potentially Wasteful

Example: A process using 4 GB RAM with very low CPU activity, no disk/network activity, and 40 minutes of inactivity may receive a high waste score.

A high waste score never means automatic termination.

PROPOSED EVALUATION

Evaluation will test accuracy, safety, and overhead

Baseline: compare DaemonGuard with static CPU/RAM threshold detection.

Test scenarios	Evaluation metrics
Active high-memory app Idle high-memory app	Detection accuracy False-positive rate
Background app Windows service	Detection time Monitoring overhead
Critical system process Simulated abnormal process	Correct protection of system processes

IMPLEMENTATION STACK

Planned stack

- C++, Windows API, WMI, and Windows Performance Counters
- HTML/CSS/JavaScript or Windows desktop UI
- JSON or SQLite
- Visual Studio and Git

Accuracy, safety, and monitoring overhead are evaluated together.

SAFETY OVERRIDE

Critical risk overrides waste.

Critical operational risk overrides a high waste score.

USER CONTROLLED

PROTECTION LOGIC

Safety overrides optimisation when the process is critical

01 Identify Windows system processes and services.

02 Check process owner and executable path.

03 Maintain a protected-process list.

04 Calculate an Operational Risk Score.

05 Disable termination for critical processes.

06 Require user or administrator confirmation.

GreenOps view: highlight avoidable CPU time, inactive memory, inefficient background usage, and possible CPU-minutes or memory-hours saved. Energy and carbon estimation are future enhancements.

PROPOSED OUTCOME

DaemonGuard aims to make optimization understandable and safe

- Working Windows resource-monitoring prototype
- Process-level resource analysis
- Explainable waste classification
- Risk-aware recommendations
- GreenOps dashboard with historical trends

NEXT HORIZON

Future work

Adaptive thresholds, machine-learning anomaly detection, energy and carbon estimation, multi-device enterprise monitoring, and automated remediation for approved applications.

UNDERSTANDABLE · SAFE · GREENOPS

Thank You

Questions?

“DaemonGuard aims to make Windows resource monitoring understandable, safe and environmentally responsible.”